Understand Ratios

Make an equivalent ratio by multiplying both amounts by the same number.

Name _____ **Date** _____

Tonight's spark

Tahira's dressing uses 1 spoon of vinegar for every 3 spoons of olive oil. She wants 4 spoons of vinegar for a block party. A friend says: add 3 to the oil too, so use 6. Will that taste right?

Make a guess first: does she need more than 6 spoons of oil, or fewer?

Vinegar (spoons)	Olive oil (spoons)
1	3
4	?

×4

Our class model: the scale factor

Try it

Work it out on paper. Write one answer on each line.

1 **Multiply both numbers in the ratio 2:3 by 4. What is the new ratio?**

A. 6:7 **B.** 8:12 **C.** 8:3 **D.** 2:12

My answer _____

2 **What is the scale factor from 3:5 to 12:20?**

A. 9 **B.** 15 **C.** 4

My answer _____

3 **Which ratio is equivalent to 1:3?**

A. 4:12 **B.** 2:4 **C.** 3:1 **D.** 1:6

My answer _____

4 **A mix uses 2 cups of flour for every 5 cups of milk. How much milk goes with 6 cups of flour?**

A. 9 **B.** 30 **C.** 10 **D.** 15

My answer _____

5 **To go from 6:9 down to 2:3, divide both numbers by what?**

A. 2 **B.** 3 **C.** 4 **D.** 6

My answer _____

6 **Are 5:8 and 15:24 equivalent?**

A. No, the numbers are all different **B.** Yes, 5 × 3 = 15 and 8 × 3 = 24

C. No, 15 is bigger than 8 **D.** Yes, because both are bigger numbers

My answer _____

Talk about it

What number did you multiply by, and did you do the same thing to both parts?

Home mission

Use dry beans as spoons. Put 1 bean for vinegar and 3 beans for oil. Ask your student to make the dressing with 4 beans of vinegar. How many oil beans? Then ask: what did you do to both piles?

If we get stuck, ask

- What number was the first quantity multiplied by?
- Now do exactly that to the other quantity.
- Can you divide both new numbers back down to the ratio you started with?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/3-1-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it